Medical Microbiology

Third stage- College of Dentistry

Lab. 14

Neisseria spp.

Prepared by:

PhD. Dheyaa Naji Hamza

M.Sc. Zainab Majid Mohammed

Learning Objectives:

After completing this lab, you should be able to:

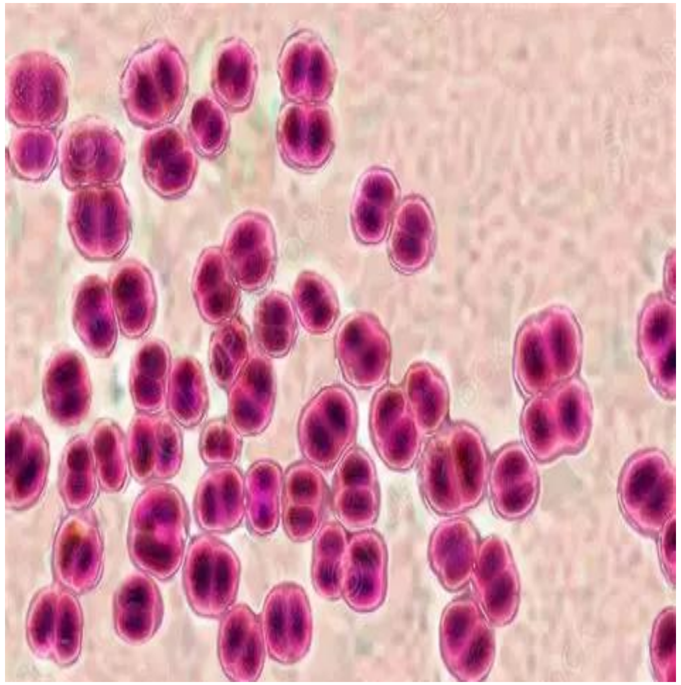
1-Describe the microscopic morphology and general Characteristics of the ***Neisseria***.

4-Discuss the pathogenic classification and major patho-types of *Neisseria*.

2-Explain the laboratory techniques used for the detection and identification of *Neisseria.*

3-Identify the cultural characteristics of *Neisseria* and the appropriate media used for its isolation.

Morphology and General Characteristics of Neisseria:



-Shaped: diplococci, typically described as **kidney-bean** or **coffee-bean** shaped with adjacent sides flattened.

-Gram stain: Gram-negative bacteria.

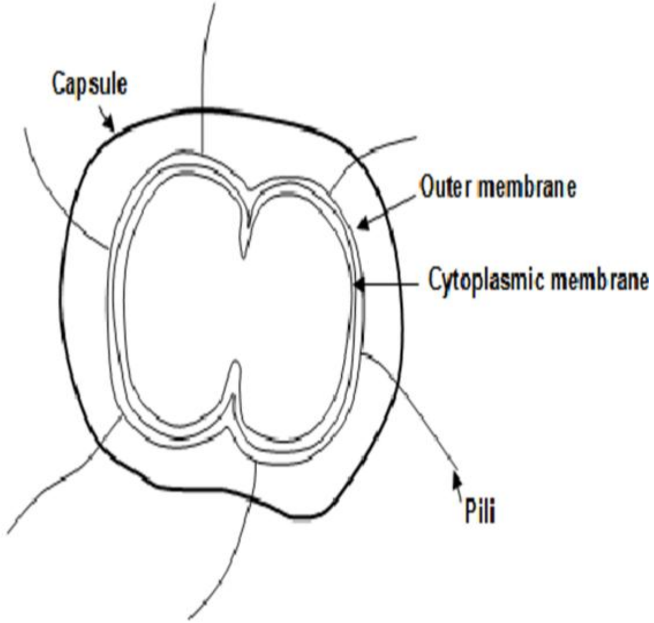
-Respiration: Aerobe and fastidious (requires enriched media) like **Chocolate Agar** for growth.

-Spore formation: non-spore forming.

-Motility: non- motile.

-Capsule:

- ❑ **Neisseria meningitidis:** capsulated, It has a polysaccharide capsule.
- ❑ **Neisseria gonorrhoeae:** non-capsulated, though some strains may have a very thin capsule-like layer.



Habitat of *Neisseria* Species:

Neisseria species colonize the **mucous membranes of humans**.

-Neisseria meningitidis

Primarily inhabits the **nasopharynx (upper respiratory tract)** and may be present in healthy asymptomatic carriers.

-Neisseria gonorrhoeae

Mainly colonizes the **urogenital tract**.

-Commensal *Neisseria*

Some species are part of the **normal oral flora** and are generally non-pathogenic, including:

- ***Neisseria sicca***
- ***Neisseria mucosa***

These organisms typically do not cause disease under normal conditions.

Pathogenic *Neisseria* Species:

There are two major pathogenic species in humans:

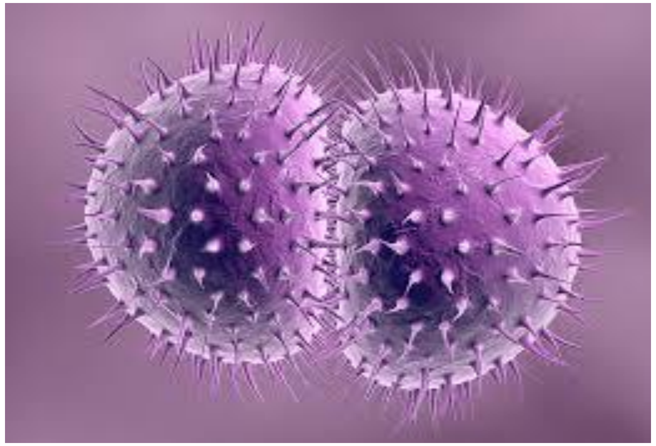
1-Neisseria gonorrhoeae (Gonococcus):

Causes **gonorrhea**, Can also lead to **gonococcal pharyngitis**, which is important in dental practice due to oropharyngeal infection.

2-Neisseria meningitidis (Meningococcus)

Causes **meningitis** and **septicemia**. Frequently found in the nasopharynx of asymptomatic carriers, facilitating transmission.

Virulence Factors:



1-Pili (Fimbriae):

Enable attachment to mucosal epithelial cells.

2-Capsule (mainly in N. meningitidis):

Polysaccharide capsule protects against phagocytosis.

3-Lipooligosaccharide (LOS):

Acts as endotoxin; triggers strong inflammatory response leading to tissue damage and septic shock.

4-IgA Protease:

Breaks down secretory IgA on mucosal surfaces, enhancing colonization.

Diagnosis Methods:

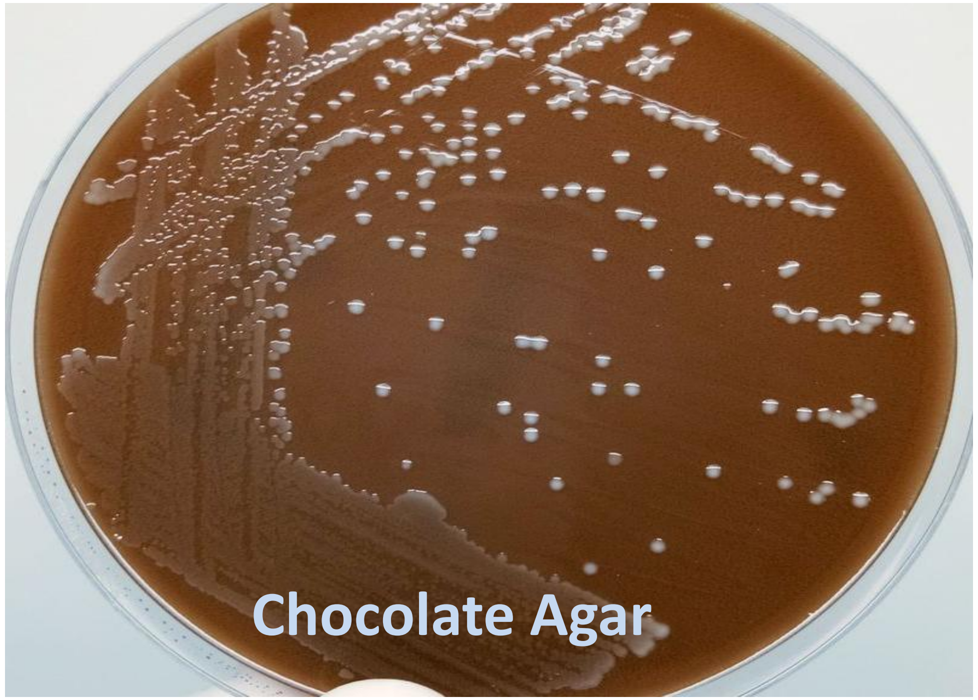
1-Culture Media:

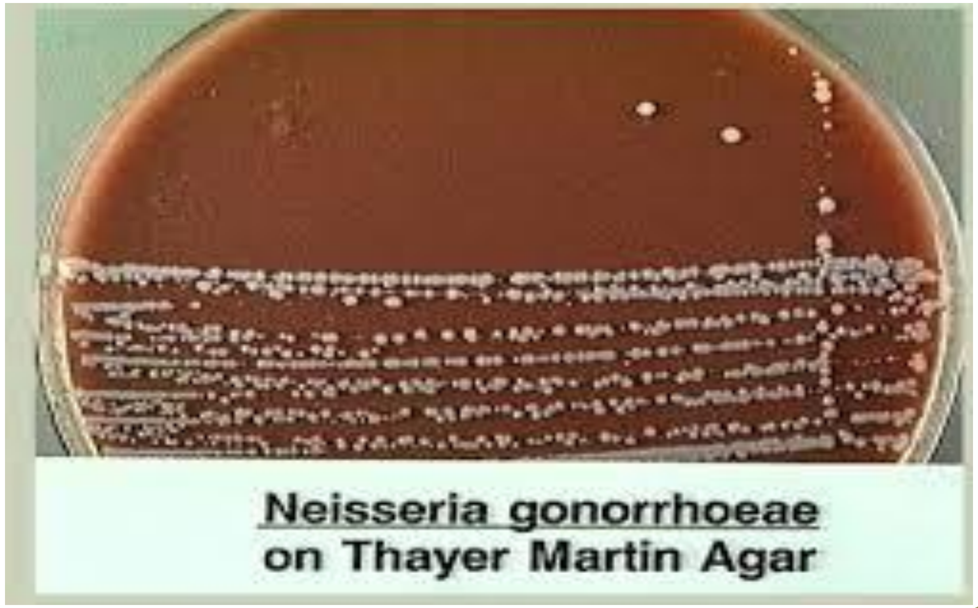
They grow best at 37°C in an atmosphere enriched with 5-10% CO_2.

A-Blood Agar: Used mainly for N. meningitidis.

B-Chocolate Agar (CHOC): A non-selective enriched medium containing heated blood, which releases essential growth factors.

C-Modified Thayer-Martin (MTM) Agar: A selective medium based on **chocolate agar** with added **antibiotics** (Vancomycin, Colistin, Nystatin, and Trimethoprim) to inhibit the growth of Gram positive, Gram negative and fungi, allowing only Neisseria species to grow.





2-Biochemical Tests

1-Oxidase Test: positive.

2-Catalase Test: positive except N. elongata.

3-Carbohydrate Utilization Test (CTA Sugars):

Principle: Neisseria species are oxidative, not fermentative. They produce weak acids when they break down carbohydrates in the presence of oxygen.

Medium: CTA (Cystine Trypticase Agar) with 1% carbohydrate and Phenol Red as a pH indicator.

Results:

Positive: The medium turns Yellow (acid production) at the top of the tube.

Negative: The medium remains Red/Orange.

Species Differentiation:

- **N. gonorrhoeae**: Oxidizes **Glucose** only.
- **N. meningitidis**: Oxidizes **Glucose** and **Maltose**.
- **N. lactamica**: Oxidizes **Glucose**, **Maltose**, and **Lactose**.

N gonorrhoeae

N meningitidis

	Glucose	Maltose	Lactose
N. Gonorrhea	+	-	-
N. Meningitides	+	+	-

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Any Question?

Thank you